

Organ transplants commonly cause immune responses that can lead to rejection of the transplanted organ. Drugs that suppress the immune response can therefore increase the probability of successful transplants and reduce the severity of symptoms associated with rejection.

Calcineurin is a protein phosphatase that activates T cells by dephosphorylating the transcription factor NFAT, and calcineurin inhibitors are frequently used in combination with organ transplants. The protein RCAN1 (Regulator of Calcineurin 1) is an endogenous noncompetitive inhibitor of calcineurin that binds at an allosteric site. The phosphorylated form, pRCAN1, cannot bind the allosteric site of calcineurin. However, pRCAN1 does bind the active site, where it acts as a substrate and becomes dephosphorylated. This converts it back to RCAN1. Interactions between calcineurin, RCAN1, and pRCAN1 are shown in Figure 1.

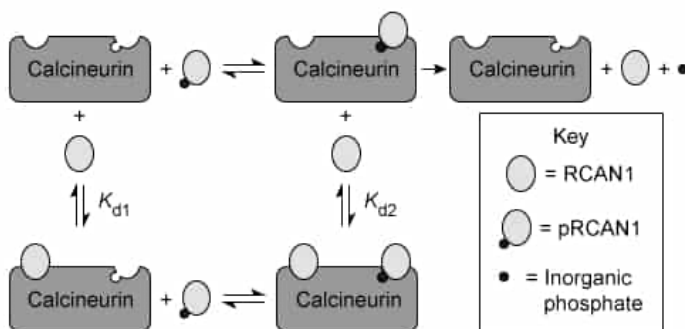


Figure 1 Schematic of calcineurin interactions with its substrate (pRCAN1) and a noncompetitive inhibitor (RCAN1)

Generation of pRCAN1 is triggered when a specific ligand binds to certain G protein-coupled receptors. These receptors then initiate a phosphorylation cascade, eventually phosphorylating the enzyme p38 α . This activates p38 α , which then transfers a phosphate group from ATP to RCAN1, yielding pRCAN1. Thus calcineurin activity may be regulated by altering levels of the ligand, which in turn alters the relative amount of unphosphorylated RCAN1 available to inhibit calcineurin.

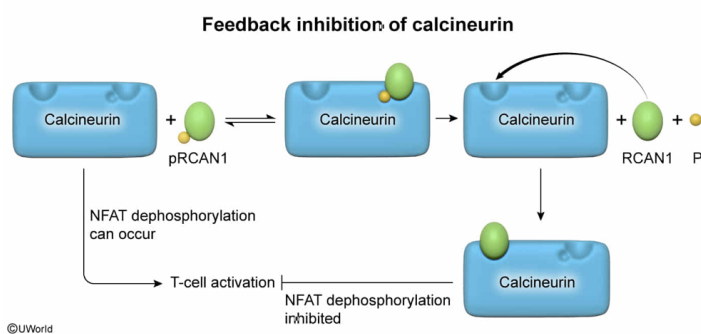
Dephosphorylation of pRCAN1 by calcineurin most likely causes which of the following?

- ☐ A. Increased calcineurin activity [16%]
- ☐ B. Upregulation of G protein-coupled receptor activity [10%]
- ☐ C. Decreased p38 α activity [15%]
- ✔ ☒ D. Downregulation of T-cell activation [57%]

Correct

58% answered correctly

Explanation:



Many enzymes participate in **feedback inhibition**, in which the **product of the reaction inhibits the enzyme** by binding to an **allosteric site**. The passage states that pRCAN1 can be dephosphorylated by calcineurin, which yields RCAN1. RCAN1 (the product) can then inhibit calcineurin.

The passage also states that

1. organ transplants are more successful when immune responses are suppressed
2. calcineurin activates T cells (a type of white blood cell involved in immune responses) by dephosphorylating NFAT, and
3. calcineurin inhibitors are commonly used in combination with organ transplant procedures.

These statements indicate that calcineurin inhibition helps suppress the immune response by decreasing T-cell activation. Dephosphorylation of pRCAN1 most likely leads to inhibition of calcineurin by producing the inhibitor RCAN1. This would result in inability to dephosphorylate NFAT, which would lead to downregulation of T-cell activation.

(Choice A) Dephosphorylation of pRCAN1 leads to increased levels of RCAN1, which *decreases* (not increases) calcineurin activity.

(Choice B) Upregulation of **G protein-coupled receptors** would lead to increased RCAN1 phosphorylation, but there is no indication that the phosphorylation state of RCAN1 affects G-protein activity. Instead, G-protein activity is controlled by levels of the relevant ligand.

(Choice C) There is no indication that pRCAN1 dephosphorylation is the *cause* of any decrease in p38 α activity. In addition, neither reducing the concentration of a product (pRCAN1) nor increasing the concentration of a substrate (RCAN1) would be expected to decrease p38 α activity. Instead, dephosphorylation of p38 α , not pRCAN1, would cause decreased p38 α activity.

Educational objective:

Feedback regulation occurs when the product of an enzymatic reaction inhibits the enzyme.

Time spent:0 sec
Subject:
Biochemistry

QID:403131
System:
Enzymes

Organ transplants commonly cause immune responses that can lead to rejection of the transplanted organ. Drugs that suppress the immune response can therefore increase the probability of successful transplants and reduce the severity of symptoms associated with rejection.

Calcineurin is a protein phosphatase that activates T cells by dephosphorylating the transcription factor NFAT, and calcineurin inhibitors are frequently used in combination with organ transplants. The protein RCAN1 (Regulator of Calcineurin 1) is an endogenous noncompetitive inhibitor of calcineurin that binds at an allosteric site. The phosphorylated form, pRCAN1, cannot bind the allosteric site of calcineurin. However, pRCAN1 does bind the active site, where it acts as a substrate and becomes dephosphorylated. This converts it back to RCAN1. Interactions between calcineurin, RCAN1, and pRCAN1 are shown in Figure 1.

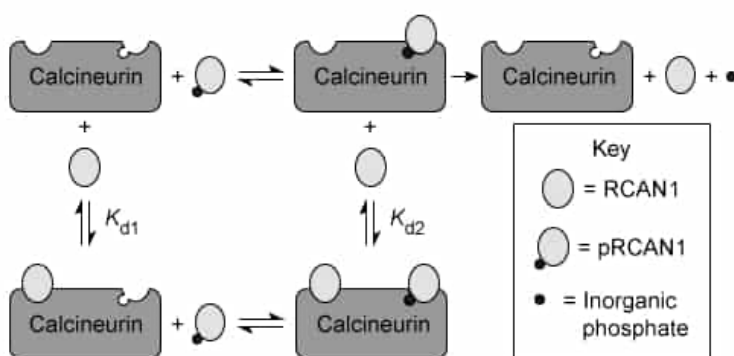


Figure 1 Schematic of calcineurin interactions with its substrate (pRCAN1) and a noncompetitive inhibitor (RCAN1)

Generation of pRCAN1 is triggered when a specific ligand binds to certain G protein-coupled receptors. These receptors then initiate a phosphorylation cascade, eventually phosphorylating the enzyme p38 α . This activates p38 α , which then transfers a phosphate group from ATP to RCAN1, yielding pRCAN1. Thus calcineurin activity may be regulated by altering levels of the ligand, which in turn alters the relative amount of unphosphorylated RCAN1 available to

inhibit calcineurin.

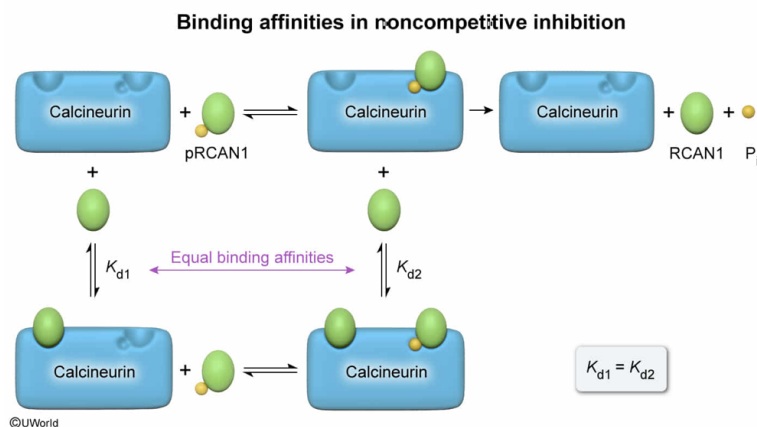
Given that RCAN1 noncompetitively inhibits calcineurin, which statement is true regarding the dissociation constants shown in Figure 1?

- ☐ A. $K_{d1} < K_{d2}$ [22%]
- ☐ B. $K_{d1} > K_{d2}$ [22%]
- ✓ ☒ C. $K_{d1} = K_{d2}$ [38%]
- ☐ D. The relationship between K_{d1} and K_{d2} cannot be determined without more information. [15%]

Correct

38% answered correctly

Explanation:



Enzyme inhibitors are classified by their relative affinities for free enzyme (E) and enzyme-substrate complex (ES):

- **Competitive inhibitors** bind E only
- **Uncompetitive inhibitors** bind ES only
- **Mixed inhibitors** bind both E and ES, though they may have a higher affinity for E than for ES, or vice versa
- **Noncompetitive inhibitors** are a special case of mixed inhibitor

in which the affinities for E and ES are exactly equal

The affinity of a ligand for a protein is measured by its **dissociation constant K_d** , where a **smaller K_d** indicates **greater binding affinity**. In this case, the ligand is the inhibitor RCAN1, and the protein is the enzyme calcineurin. Based on Figure 1, K_{d1} is the dissociation constant for the interaction between RCAN1 and free calcineurin (E), and K_{d2} is the dissociation constant for the interaction between RCAN1 and the calcineurin-pRCAN1 complex (ES).

The passage and question state that RCAN1 **noncompetitively inhibits** calcineurin. Therefore, RCAN1 must have **equal affinity for free calcineurin and the calcineurin-pRCAN1 complex**. As a result, the two **K_d values must be equal**: $K_{d1} = K_{d2}$.

(Choices A and B) If K_{d1} were less than K_{d2} , this would indicate a mixed inhibitor with a higher affinity for free calcineurin than for the calcineurin-pRCAN1 complex. Conversely, if K_{d1} were greater than K_{d2} , RCAN1 would have a higher affinity for the complex than for free enzyme. To be a noncompetitive inhibitor, the affinities (and K_d values) must be equal.

(Choice D) The information that RCAN1 is a noncompetitive inhibitor is sufficient to determine that $K_{d1} = K_{d2}$.

Educational objective:

Noncompetitive inhibitors bind free enzyme and enzyme-substrate complex with equal affinity. The affinity of any ligand for any protein (including inhibitor-enzyme interactions) can be measured by the dissociation constant K_d , where a smaller K_d indicates a higher binding affinity.

Time spent:0 sec

Subject:
Biochemistry

QID:403132

System:
Enzymes

Organ transplants commonly cause immune responses that can lead to rejection of the transplanted organ. Drugs that suppress the immune response can therefore increase the probability of successful transplants and reduce the severity of symptoms associated with rejection.

Calcineurin is a protein phosphatase that activates T cells by dephosphorylating the transcription factor NFAT, and calcineurin inhibitors are frequently used in combination with organ transplants. The protein RCAN1 (Regulator of Calcineurin 1) is an endogenous noncompetitive inhibitor of calcineurin that binds at an allosteric site. The phosphorylated form, pRCAN1, cannot bind the allosteric site of calcineurin. However, pRCAN1 does bind the active site, where it acts as a substrate and becomes dephosphorylated. This converts it back to RCAN1. Interactions between calcineurin, RCAN1, and pRCAN1 are shown in Figure 1.

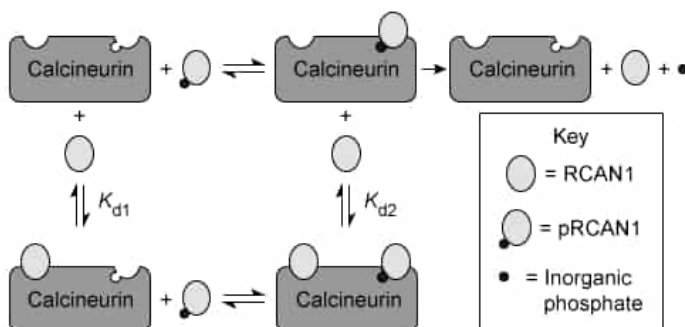


Figure 1 Schematic of calcineurin interactions with its substrate (pRCAN1) and a noncompetitive inhibitor (RCAN1)

Generation of pRCAN1 is triggered when a specific ligand binds to certain G protein-coupled receptors. These receptors then initiate a phosphorylation cascade, eventually phosphorylating the enzyme p38 α . This activates p38 α , which then transfers a phosphate group from ATP to RCAN1, yielding pRCAN1. Thus calcineurin activity may be regulated by altering levels of the ligand, which in turn alters the relative amount of unphosphorylated RCAN1 available to inhibit calcineurin.

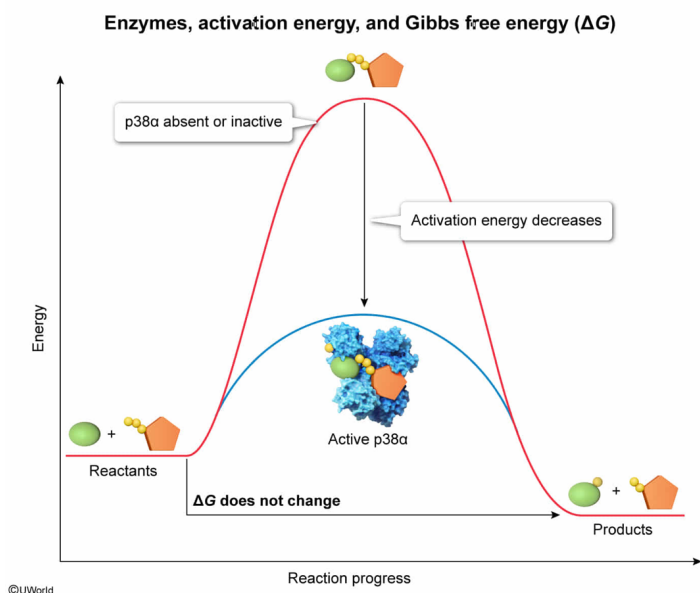
How is the energy of the phosphate transfer from ATP to RCAN1 affected by the addition of active p38 α ?

- ✔ ☒ A. The activation energy decreases but the Gibbs free energy is unaffected. [77%]
- ☐ B. The activation energy increases but the Gibbs free energy decreases. [5%]
- ☐ C. The activation energy increases but the Gibbs free energy is unaffected. [6%]
- ☐ D. The activation energy and the Gibbs free energy both decrease. [9%]

Correct

77% answered correctly

Explanation:



Enzymes are biological catalysts. **Catalysts** in general are substances that **increase the rate** of a chemical reaction **without being consumed** by the reaction. All catalysts, including enzymes, work by **decreasing** the **activation energy** of a reaction. However, because **Gibbs free energy ΔG** depends only on the identities of the products and reactants, and not on the mechanism of the reaction (ie, ΔG is a **function of state**), **catalysts never affect the ΔG** of any reaction.

The passage states that p38α is an enzyme that transfers a phosphate group from ATP to RCAN1 (ie, p38α is a **kinase**). Because p38α is an

enzyme, it must **decrease the activation energy** of the reaction, allowing the reaction to proceed faster. However, the reactants are ATP and RCAN1, and the products are ADP and pRCAN1, *regardless* of whether p38 α is present. Therefore, as with any catalyst, p38 α **does not affect the Gibbs free energy** of the reaction.

(Choices B and D) Enzymes never decrease (or increase) the Gibbs free energy of a reaction.

(Choice C) Although it is true that enzymes do not affect the Gibbs free energy of the reaction they catalyze, enzymes cause a *decrease* in activation energy, not an increase.

Educational objective:

Enzymes are biological catalysts that increase the rate of a biological reaction without being consumed. As such, they decrease the activation energy of the reaction they catalyze but do not alter the Gibbs free energy.

Time spent:0 sec
Subject:
Biochemistry

QID:403133
System:
Enzymes

Organ transplants commonly cause immune responses that can lead to rejection of the transplanted organ. Drugs that suppress the immune response can therefore increase the probability of successful transplants and reduce the severity of symptoms associated with rejection.

Calcineurin is a protein phosphatase that activates T cells by dephosphorylating the transcription factor NFAT, and calcineurin inhibitors are frequently used in combination with organ transplants. The protein RCAN1 (Regulator of Calcineurin 1) is an endogenous noncompetitive inhibitor of calcineurin that binds at an allosteric site. The phosphorylated form, pRCAN1, cannot bind the allosteric site of calcineurin. However, pRCAN1 does bind the active site, where it acts as a substrate and becomes dephosphorylated. This converts it back to RCAN1. Interactions between calcineurin, RCAN1, and pRCAN1 are shown in Figure 1.

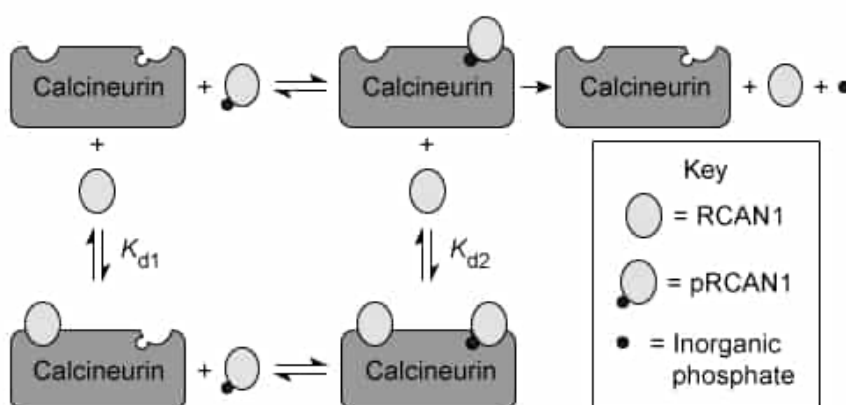


Figure 1 Schematic of calcineurin interactions with its substrate (pRCAN1) and a noncompetitive inhibitor (RCAN1)

Generation of pRCAN1 is triggered when a specific ligand binds to certain G protein-coupled receptors. These receptors then initiate a phosphorylation cascade, eventually phosphorylating the enzyme p38 α . This activates p38 α , which then transfers a phosphate group from ATP to RCAN1, yielding pRCAN1. Thus calcineurin activity may be regulated by altering levels of the ligand, which in turn alters the relative amount of unphosphorylated RCAN1 available to inhibit calcineurin.

Based on the information in the passage, which of the following best describes the regulation of p38 α activity?

- ✔ ☒ A. p38 α is activated by a covalent modification.
[78%]
- ☐ B. p38 α is inactivated by a covalent modification.
[2%]
- ☐ C. p38 α is activated by a noncovalent modification.
[18%]
- ☐ D. p38 α is inactivated by a noncovalent modification.

Correct

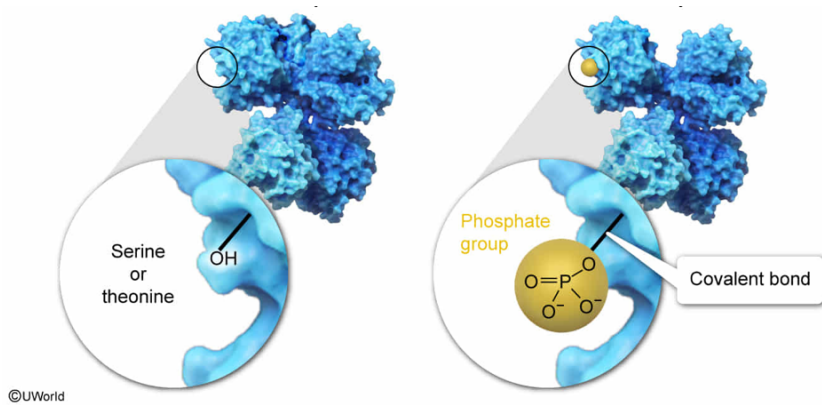
77% answered correctly

Explanation:

Covalent modifications alter enzyme activity

Inactive p38 α

Active p38 α



Enzyme activity may be **modulated by covalent modifications** to the enzyme (ie, **post-translational modifications**) such as phosphorylation or glycosylation. Whether a covalent modification activates or inactivates an enzyme varies from one enzyme to another, and even from one modification to another within an enzyme. For example, phosphorylation at one amino acid residue may activate the enzyme, while phosphorylation at another may inactivate it (or decrease its activity).

The passage states that p38 α becomes active when phosphorylated. Therefore, p38 α activity is regulated by a covalent modification that activates the enzyme.

(Choice B) p38 α is activated, not inactivated, by the covalent addition of a phosphate group.

(Choices C and D) Phosphorylation is a covalent modification in which a phosphate group forms a **covalent bond** with an amino acid residue. Noncovalent modifications that alter enzyme activity occur when ligands bind to **allosteric sites**. Although the passage indicates noncovalent regulation of *calcineurin*, it mentions only *covalent* regulation of p38 α .

Educational objective:

Enzyme activity may be modulated by formation of covalent bonds between amino acid residues and various chemical groups, such as phosphates (phosphorylation) or sugars (glycosylation). These modifications may increase or decrease the activity of the enzyme, depending on the enzyme itself and the site of the modification.

Time spent:0 sec**Subject:**

Biochemistry

QID:403134**System:**

Enzymes